KING'S OWN INSTITUTE*

Success in Higher Education

ICT371 ARTIFICIAL INTELLIGENCE T126

All information in the Subject Outline is correct at the time of approval. KOI reserves the right to make changes to the Subject Outline if they become necessary. Any changes require the approval of the KOI Academic Board and will be formally advised to those students who may be affected by email and via Moodle. Information contained within this Subject Outline applies to students enrolled in the trimester as indicated

1. General Information

1.1 Administrative Details

Associated HE Award(s)	Duration	Level	Subject Coordinator
Bachelor of Information Technology (BIT)	1 trimester	Level 3	Saeid Iranmanesh saeid.iranmanesh@koi.edu.au P: +61 (2) 9283 3583 L: 7-11,11 York St. Consultation: via Moodle or by appointment.

1.2 Core / Elective

Elective subject for BIT

1.3 Subject Weighting

Indicated below is the weighting of this subject and the total course points.

Subject Credit Points	Total Course Credit Points
4	BIT (96 Credit Points)

1.4 Student Workload

Indicated below is the expected student workload per week for this subject

No. Timetabled Hours/Week*	No. Personal Study Hours/Week**	Total Workload Hours/Week***
4 hours/week (2 hour Lecture + 2 hour Tutorial)	6 hours/week	10 hours/week

* Total time spent per week at lectures and tutorials
** Total time students are expected to spend per week in studying, completing assignments, etc.
*** Combination of timetable hours and personal study.

1.1 Mode of Delivery:

Classes will be face-to-face or hybrid. Certain classes will be online (e.g., special arrangements).

1.6 Pre-requisites:

ICT104 Program Design and Development and Successful completion of 48 credit points

1.7 General Study and Resource Requirements

- Dedicated computer laboratories are available for student use. Normally, tutorial classes are conducted in the computer laboratories

- Students are expected to attend classes with the requisite textbook and must read specific chapters prior to each tutorial. This will allow them to actively take part in discussions. Students should have elementary skills in both word processing and electronic spreadsheet software, such as Office 365 or MS Office
- Computers and WIFI facilities are extensively available for student use throughout KOI. Students are encouraged to make use of the campus Library for reference materials
- Students will require access to the internet and email. Where students use their own computers, they should have internet access. KOI will provide access to required software.

Resource requirements specific to this subject: Rapid miner, Office 365, MS Imagine.

1.8 Academic Advising

Academic advising is available to students throughout teaching periods including the exam weeks. As well as requesting help during scheduled class times, students have the following options:

- Consultation times: A list of consultation hours is provided on the homepage of Moodle where appointments can be booked.
- Subject coordinator: Subject coordinators are available for contact via email. The email address of the subject coordinator is provided at the top of this subject outline.
- Academic staff: Lecturers and Tutors provide their contact details in Moodle for the specific subject. In most cases, this will be via email. Some subjects may also provide a discussion forum where questions can be raised.
- Head of Program: The Head of Program is available to all students in the program if they need advice about their studies and KOI procedures.
- Vice President (Academic): The Vice President (Academic) will assist students to resolve complex issues (but may refer students to the relevant lecturers for detailed academic advice).

2 Academic Details

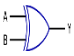
2.1 Overview of the Subject

The goal of Artificial Intelligence is to build software systems that behave "intelligently". That is, do these computer systems "do the right thing" in complex environments? Do they act optimally given the limited information and computational resources available? How is this aim interpreted? This subject covers the core topics of Artificial Intelligence such as knowledge representation, reasoning, and learning. Students will learn to design and analyse autonomous agents that do the right thing in the face of limited computational resources and limited information. This subject examines agents that can effectively make decisions in fully observable, partially observable and adversarial environments, and agents that can adapt their actions by learning from experience.

2.2 Graduate Attributes for Undergraduate Courses

Graduates of Bachelor courses from King's Own Institute (KOI) will achieve the graduate attributes expected under the Australian Qualifications Framework (2nd edition, January 2013). Graduates at this level will be able to apply a broad and coherent body of knowledge from their major area of study in a range of contexts for professional practice or scholarship and as a pathway for further learning.

King's Own Institute's generic graduate attributes for a bachelor's level degree are summarised below:

	KOI Bachelor Degree Graduate Attributes	Detailed Description
	Knowledge	Current, comprehensive, and coherent and connected knowledge
	Critical Thinking	Critical thinking and creative skills to analyse and synthesise information and evaluate new problems
	Communication	Communication skills for effective reading, writing, listening and presenting in varied modes and contexts and for transferring knowledge and skills to a variety of audiences
	Information Literacy	Information and technological skills for accessing, evaluating, managing and using information professionally
	Problem Solving Skills	Skills to apply logical and creative thinking to solve problems and evaluate solutions
	Ethical and Cultural Sensitivity	Appreciation of ethical principles, cultural sensitivity and social responsibility, both personally and professionally
	Teamwork	Leadership and teamwork skills to collaborate, inspire colleagues and manage responsibly with positive results
	Professional Skills	Professional skills to exercise judgement in planning, problem solving and decision making

Across the course, these skills are developed progressively at three levels:

- **Level 1 Foundation** – Students learn the basic skills, theories and techniques of the subject and apply them in basic, standalone contexts
- **Level 2 Intermediate** – Students further develop the skills, theories and techniques of the subject and apply them in more complex contexts, and begin to integrate this application with other subjects.
- **Level 3 Advanced** – Students demonstrate an ability to plan, research and apply the skills, theories and techniques of the subject in complex situations, integrating the subject content with a range of other subject disciplines within the context of the course.

2.3 Subject Learning Outcomes

This is a Level 3 subject.

On successful completion of this subject, students should be able to:

Subject Learning Outcomes	Contribution to Graduate Attributes
a) Identify problems that are amenable to solution by AI methods and select AI methods suited to solving such problems	
b) Formulate a given problem in the language/framework of different AI methods (for example, as a search problem, as	

a constraint satisfaction problem, or as a planning problem)	
c) Analyse AI algorithms (e.g., standard search or constraint propagation algorithms)	
d) Solve simple problems using AI techniques and algorithms.	

2.4 Subject Content and Structure

Below are details of the subject content and how it is structured, including specific topics covered in lectures and tutorials. Reading refers to the text unless otherwise indicated.

Weekly Planner:

Week (beginning)	**Topic covered in each week's lecture**	**Reading(s)**	**Expected work as listed in Moodle**
Week 1 02 March	Introduction to AI	Ch. 1	Tutorial exercises based on lecture topics are intended to stimulate discussion. Discussing AI applications and understanding the latest AI technologies. Formative weekly tutorial
Week 2 09 March	Intelligent Agents	Ch. 2	Tutorial exercises based on lecture topics are intended to stimulate discussion. Introduction to rapid miner as an AI tool. Summative worth 1%
Week 3 16 March	Solving Problem by Searching	Ch. 3	Tutorial exercises based on lecture topics are intended to stimulate discussion. Rapid Miner: Data loading and problem understanding, problem space understanding. Summative worth 1%
Week 4 23 March	Beyond Classical Search	Ch. 4	Tutorial exercises based on lecture topics are intended to stimulate discussion. Discussing and analysing genetic algorithms in terms of cross-over, mutation, parent selection, local and absolute minima/maxima. Summative worth 1% **Assessment 2 due: Formative Quiz**

Week (beginning)	Topic covered in each week's lecture	Reading(s)	Expected work as listed in Moodle
Week 5 30 March	Adversarial search	Ch. 5	Tutorial exercises based on lecture topics are intended to stimulate discussion. Rapid Miner: decision tree Summative worth 1% **Assessment 3 due:** Intelligent Agents analysis - Individual
Week 6 07 April (Tue)	Constraint satisfaction problem	Ch. 6	Tutorial exercises based on lecture topics are intended to stimulate discussion. Rapid Miner: artificial neural network. Summative worth 1%
Week 7 13 April	Logical Agent	Ch. 7	Tutorial exercises based on lecture topics are intended to stimulate discussion. Rapid Miner: K-means Summative worth 1%
Week 8 20 April	First order logic	Ch. 8	Tutorial exercises based on lecture topics are intended to stimulate discussion. Discussing logic programming to build a solver. Summative worth 1% **Assessment 4 due:** Search Strategy Project - Individual Report
Week 9 28 April (Tue)	Probability reasoning	Ch. 14	Tutorial exercises based on lecture topics are intended to stimulate discussion. Rapid Miner: text mining. Summative worth 1% **Assessment 4 due:** Search Strategy Project - Individual Demonstration
Week 10 04 May	Learning from examples 1	Ch. 18	Tutorial exercises based on lecture topics are intended to stimulate discussion. Rapid Miner: Association Rules Summative worth 1%
Week 11 11 May	Learning from examples 2	Ch. 18	Tutorial exercises based on lecture topics are intended to stimulate discussion. Rapid Miner: Some Tips for Assignment 2 Summative worth 1%

Week (beginning)	Topic covered in each week's lecture	Reading(s)	Expected work as listed in Moodle
			Assessment 5 due: Knowledge based Project-group - Report Submission.
Week 12 18 May	Revision	All Chapters	Revision **Assessment 5 due:** Knowledge based Project-group - Demonstration.
Week 13 25 May	Study review week and Final Exam Week		
Week 14 01 June	Examinations Continuing students - enrolments for T226 open		Please see exam timetable for exam date, time and location
Week 15 09 June (Tue)	Student Vacation begins New students - enrolments for T226 open		
Week 16 15 June	• Results Released • Review of Grade Day for T126 – see Sections 2.6 and 3.2 below for relevant information. • Certification of Grades NOTE: More information about the dates will be provided at a later date through Moodle/KOI email.		
T226 29 June 2026			
Week 1 29 June	Week 1 of classes for T226		

2.5 Public Holiday Amendments

Please note: KOI is closed on all scheduled NSW Public Holidays.

T126 has Five (5) public holidays that occur during this trimester. Classes scheduled for these public holidays (Calendar Class Date) will be rescheduled as per the table below.

This applies to ALL subjects taught in T126.

Please see the table below and adjust your class timing as required. Please make sure you have arrangements in place to attend the rescheduled classes if applicable to your T126 enrolment.

Classes will be conducted at the same time and in the same location as your normally scheduled class except these classes will be held on the date shown below.

Calendar Class Date	Rescheduled Class Date
03 April 2026 (Friday) 04 April 2026 (Saturday) 06 April 2026 (Monday) 25 April 2026 (Saturday) 27 April 2026 (Monday)	25 May 2026 (Monday) 26 May 2026 (Tuesday) 27 May 2026 (Wednesday)

2.6 Review of Grade, Deferred Exams & Supplementary Exams/Assessments

Review of Grade:

There may be instances when you believe that your final grade in a subject does not accurately reflect your performance against the marking criteria. Section 8 of the *Assessment and Assessment Appeals Policy* (www.koi.edu.au) describes the grounds on which you may apply for a Review of Grade.

If you have a concern about your marks and you are unable to resolve it with the Academic staff concerned, then you can apply for a formal Review of Grade as explained in section 3.2(e) Appeals Process below. Please note the time limits for requesting a review. Please ensure you read the Review of Grade information before applying.

Review of Grade Day:

Final exam scripts will not normally be returned to students. Students can obtain feedback on their exam performance and their results for the whole subject at the Review of Grade Day. KOI will hold the Review of Grade Day for all subjects studied in T126. **The ROG day will be announced at a later date and the students will be notified through Moodle/KOI email.**

Only final exams and whole subject results will be discussed as all other assessments should have been reviewed during the trimester. Further information about Review of Grade Day will be available through Moodle.

If you fail one or more subjects and you wish to consider applying for a Review of Grade you are *STRONGLY ADVISED to* attend the Review of Grade Day. You will have the chance to discuss your final exam and subject result with your lecturer, and will be advised if you have valid reasons for applying for a Review of Grade (see Section 3.2 below and the *Assessment and Assessment Appeals Policy*).

A formal request for a review of grade may not be considered unless you first contact the subject coordinator to discuss the result.

Deferred Exams:

If you wish to apply for a deferred exam because you are unable to attend the scheduled exam, you should submit the Assignment Extension / Exam Deferment Form available by contacting academic@koi.edu.au as soon as possible*, but no later than three (3) working days of the assessment due date.*

If you miss your mid-trimester or final exam there is no guarantee you will be offered a deferred exam.

You must apply within the stated timeframe and satisfy the conditions for approval to be offered a deferred exam (see Section 8.1 of the *Assessment and Assessment Appeals Policy and the Application for Assignment Extension or Deferred Exam Forms*). In assessing your request for a deferred exam, KOI will take into account the information you provide, the severity of the event or circumstance, your performance on other items of assessment in the subject, class attendance and your history of previous applications for special consideration.

Deferred mid-trimester exams will be held before the end of week 9. Deferred final exams will be held on two days during week 1 or 2 in the next trimester. You will not normally be granted a deferred exam on the grounds that you mistook the time, date or place of an examination, or that you have made arrangements to be elsewhere at that time; for example, have booked plane tickets.

If you are offered a deferred exam, but do not attend *you will be awarded 0 marks for the exam*. This may mean it becomes difficult for you to pass the subject. If you apply for a deferred exam within the required time frame and satisfy the conditions you will be advised by email (to your KOI student email address) of the time and date for the deferred exam. Please ensure that you are available to take the exam at this time.

Marks awarded for the deferred exam will be the marks awarded for that item of assessment towards your final mark in the subject.

Supplementary Assessments (Exams and Assessments):

A supplementary assessment may be offered to students to provide a final opportunity to demonstrate successful achievement of the learning outcomes of a subject. *Supplementary assessments are only offered at the discretion of the Board of Examiners.* In considering whether or not to offer a supplementary assessment, KOI will take into account your performance on all the major assessment items in the subject, your attendance, participation and your history of any previous special considerations.

If you are offered a supplementary assessment, you will be advised by email to your *KOI student email address* of the time and due date for the supplementary assessment – supplementary exams will normally be held at the same time as deferred final exams during week 1 or week 2 of the next trimester.

You must pass the supplementary assessment to pass the subject. The maximum grade you can achieve in a subject based on a supplementary assessment is a PASS grade.

If you:

- are offered a supplementary assessment, but fail it;
- are offered a supplementary exam, but do not attend; or
- are offered a supplementary assessment but do not submit by the due date; you will receive a FAIL grade for the subject.

Students are also eligible for a supplementary assessment for their final subject in a course where they fail the subject but have successfully completed all other subjects in the course. You must have completed all major assessment tasks for the subject and obtained a passing mark on at least one of the major assessment tasks to be eligible for a supplementary assessment.

If you believe you meet the criteria for a supplementary assessment for the final subject in your course, but have not received an offer, complete the *Complaint, Grievance, Appeal Form* and send your form to reception@koi.edu.au. The deadline for applying for supplementary assessment is the Friday of the first week of classes in the next trimester.

2.7 Teaching Methods/Strategies

Briefly described below are the teaching methods/strategies used in this subject:

- *Lectures* (2 hours/week) are conducted in seminar style and address the subject content, provide motivation and context and draw on the students' experience and preparatory reading.
- *Tutorials* (2 hours/week) include class discussion of case studies and research papers, practice sets and problem-solving and syndicate work on group projects. Tutorials often include group exercises and so contribute to the development of teamwork skills and cultural understanding. Tutorial participation is an essential component of the subject and contributes to the development of many of the graduate attributes (see section 2.2 above). Tutorial participation contributes towards the assessment in many subjects (see details in Section 3.1 for this subject). Supplementary tutorial material such as case studies, recommended readings, review questions etc. will be made available each week in Moodle.

- *Online* teaching resources include class materials, readings, model answers to assignments and exercises and discussion boards. All online materials for this subject as provided by KOI will be found in the Moodle page for this subject. Students should access Moodle regularly as material may be updated at any time during the trimester
- *Other contact* - academic staff may also contact students either via Moodle messaging, or via email to the email address provided to KOI on enrolment.

2.8 Student Assessment

Assessment is designed to encourage effective student learning and enable students to develop and demonstrate the skills and knowledge identified in the subject learning outcomes. Assessment tasks during the first half of the study period are usually intended to maximise the developmental function of assessment (formative assessment). These assessment tasks include weekly tutorial exercises (as indicated in the weekly planner) and low stakes graded assessment (as shown in the graded assessment table). The major assessment tasks where students demonstrate their knowledge and skills (summative assessment) generally occur later in the study period. These are the major graded assessment items shown in the graded assessment table.

Final grades are awarded by the Board of Examiners in accordance with KOI's Assessment and Assessment Appeals Policy. The definitions and guidelines for the awarding of final grades within the BIT degree are:

- HD High distinction (85-100%) an outstanding level of achievement in relation to the assessment process.
- DI Distinction (75-84%) a high level of achievement in relation to the assessment process.
- CR Credit (65-74%) a better than satisfactory level of achievement in relation to the assessment process.
- P Pass (50-64%) a satisfactory level of achievement in relation to the assessment process.
- F Fail (0-49%) an unsatisfactory level of achievement in relation to the assessment process.

Provided below is a schedule of formal assessment tasks and major examinations for the subject.

Assessment Type	When assessed	Weighting	Learning Outcomes Assessed
Assessment 1: Weekly Tutorials - Individual	Week 2 – Week 11	10%	a, b, c
Assessment 2: Formative Quiz	Week 4	0%	a
Assessment 3: Intelligent Agent Analysis - Individual	Week 5 - Report Submission	20%	a,b
Assessment 4: Search Strategy Project - individual	Week 8: Report Submission Week 9: Demonstration **In Class demonstration**	30%	a, b, c, d
Assessment 5: Knowledge-based Project - group (Report and Demonstration)	Week 11: Report submission Week 12 Demonstration **In Class demonstration**	40% (30% report + 10% demonstration)	a, b, c, d

Requirements to Pass the Subject:

To gain a pass or better in this subject, students must gain a *minimum of 50%* of the total available subject marks.

2.9 Prescribed and Recommended Readings

Provided below, in formal reference format, is a list of the prescribed and recommended readings.

Prescribed Text:

Russell, S, & Norvig, P 2021, *Artificial Intelligence: A Modern Approach*, Global 4th edition, Pearson Education Limited, Harlow, United Kingdom. ISBN: 9780134610993

Recommended Readings:

Huda, N.U., Ahmed, I., Adnan, M., Ali, M., and Naeem, F., 2024. Experts and intelligent systems for smart homes' Transformation to Sustainable Smart Cities: *A comprehensive review*. Expert Systems with Applications, 238, p.122-380. DOI: 10.1016/j.eswa.2023.122380

Hejase, M., Katis, A., and Mavridou, A., 2024. Design, formalization, and verification of decision making for intelligent systems. In: AIAA SCITECH 2024 Forum, p.2409. DOI: 10.2514/6.2024-2409

Ahmed, H.M. and Sorour, S.E., 2024. Classification-driven intelligent system for automated evaluation of higher education exam paper quality. *Education and Information Technologies*, pp.1-27.
DOI: 10.1007/s10639-024-12555-9

Varsha, P.S., 2023. How can we manage biases in artificial intelligence systems–A systematic literature review. *International Journal of Information Management Data Insights*, 3(1), p.100165.
DOI: 10.1016/j.jjimei.2023.100165

Ahmad, K., Abdelrazek, M., Arora, C., Bano, M., and Grundy, J., 2023. Requirements engineering for artificial intelligence systems: A systematic mapping study. *Information and Software Technology*, 158, p.107176. DOI: 10.1016/j.infsof.2023.107176

Vyas, B., 2023. Java-Powered AI: Implementing Intelligent Systems with Code. *Journal of Science & Technology*, 4(6), pp.1-12. DOI: 10.55662/JST.2023.4601

Pimenov, D.Y., Bustillo, A., Wojciechowski, S., Sharma, V.S., Gupta, M.K. and Kuntoğlu, M., 2023. Artificial intelligence systems for tool condition monitoring in machining: Analysis and critical review. *Journal of Intelligent Manufacturing*, 34(5), pp.2079-2121. DOI: 10.1007/s10845-022-01923-2

Hu, H., Xu, J., Liu, M., and Lim, M.K., 2023. Vaccine supply chain management: An intelligent system utilizing blockchain, IoT and machine learning. *Journal of business research*, 156, p.113480.
DOI: 10.1016/j.jbusres.2022.113480

Chowdhary, K.R., 2020. *Fundamentals of Artificial Intelligence*, Springer. DOI: 10.1007/978-81-322-3972-7

Sutton, R.S., and Barto, A.G., 2018, *Reinforcement learning: an introduction*, Second edition, The MIT Press, Cambridge, Massachusetts. ISBN: 9780262039246

Recommended Journal Articles:

Jimenez, A.F., Cardenas, P.F., Canales, A., Jimenez, F., and Portacio, A., 2020. A survey on intelligent agents and multi-agents for irrigation scheduling. *Computers and Electronics in Agriculture*, 176, p.105-474. DOI: 10.1016/j.compag.2020.105474

Azami, P., Jan, T., Iranmanesh, S., Ameri Sianaki, O., and Hajiebrahimi, S., 2020. Determining the optimal restricted driving zone using genetic algorithm in a smart city. *Sensors (Basel, Switzerland)*, 20(8), p.2276. DOI:10.3390/s20082276.

Dash, B., and Sharma, P., 2022. Role of artificial intelligence in smart cities for information gathering and dissemination (a review). *Academic Journal of Research and Scientific Publishing*, 4(39). DOI:10.52132/Ajrsp.e.2022.39.4

Islam, M.R., Ahmed, M.U., Barua, S. & Begum, S., 2022. A systematic review of explainable artificial intelligence in terms of different application domains and tasks. *Applied Sciences,* 12(3), p.1353. DOI: 10.3390/app12031353

Javaid, M., Haleem, A., Singh, R.P. & Suman, R., 2022. Artificial intelligence applications for industry 4.0: A literature-based study. *Journal of Industrial Integration and Management*, 7(01), pp.83–111. DOI: 10.1142/S2424862221300040

Pallathadka, H., Ramirez-Asis, E.H., Loli-Poma, T.P., Kaliyaperumal, K., Ventayen, R.J.M. & Naved, M., 2023. Applications of artificial intelligence in business management, e-commerce and finance. *Materials Today: Proceedings*, 80, pp.2610–2613. DOI: 10.1016/j.jksuci.2024.102015

Useful Websites:

The following industry websites are useful introductory sources covering a range of information useful for this subject.

- https://www.rayven.io/
- https://aimagazine.com/
- https://www.analyticsinsight.net/magazine/
- https://news.mit.edu/topic/artificial-intelligence2

Python Resources:

https://www.python.org/ (Python Software Foundations)

Conference/ Journal Articles:

- Artificial Intelligence and Image Processing
- Engineering Intelligent Systems
- Pattern Recognition

Students are encouraged to read peer-reviewed journal articles and conference papers. Google Scholar provides a simple way to broadly search for scholarly literature. From one place, you can search across many disciplines and sources: articles, theses, books, abstracts and court opinions, from academic publishers, professional societies, online repositories, universities and other web sites.

3. Assessment Details

3.1 Details of Each Assessment Item

The assessments for this subject are described below. The description includes the type of assessment, its purpose, weighting, due date and submission requirements, the topic of the assessment, details of the task and detailed marking criteria, including a marking rubric for essays, reports and presentations. Supplementary assessment information and assistance can be found in Moodle.

KOI expects students to submit their own original work in both assignments and exams, or the original work of their group in the case of group assignments. Please retain a photocopy and softcopy of all assessments.

Marking guides for assessments follow the assessment descriptions. Students should compare final drafts of their assessment against the marking guide before submission.

Assessment 1

Assessment type: Weekly tutorials. Weeks 2 - 11

Assessment purpose: Weekly tutorial participation is designed to encourage engagement, and to develop and reinforce the knowledge and skills presented in the lectures. This assessment contributes to learning outcomes a, b, c and d.

Value: 10% **Due Date:** Weekly (1% per week)

Assessment topic: Introduction to RapidMiner Studio, importing data, data pre-processing, features extraction, process and operators in RapidMiner, Classifiers, Clustering, Twitter analysis, text processing, etc.,

Task details: Students must complete the weekly tutorial exercises and upload the answers to Moodle. Tutors will provide the feedback to the students during the activities conducted in tutorials.

Assessment 2

Assessment type: Formative - Multiple Choice Quiz – individual assignment closed book.

Purpose: This assessment will allow students to demonstrate their understanding of the topics discussed during tutorials. This assessment contributes to learning outcome a.

Value: 0% **Due Date:** Week 4 in usual tutorial timeslots

Submission requirements details: Quiz will be conducted **in class** and completed online via Moodle.

- Attend their enrolled class **in person** to do the quiz, unless their enrollment allows online attendance.
- Students with online enrollment must ensure their computer has a working **safe exam browser** (SEB) and working camera.

Task Details: The quiz will consist of a series of multiple-choice questions relating to subject content taught in weeks 1 – 3 inclusive.

Marking Rubric Quiz on Moodle:

Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Number of correct answers	Less than 5	5 to 6	6.5 to 7	7.5 to 8	8.5 to 10
Total Mark: **0 %**					

Assessment 3

Assessment Type: Written Individual Assignment (1500 Words)

Purpose: The primary objective of this assignment is to learn about the architecture of intelligent agents and how to analyse the environments in which they operate. The secondary objective is to gain an understanding of the ethical issues related to the use of intelligent agents. This assessment contributes to the learning outcomes a, b.

Value: 20% **Due Date:** Weeks 5

Submission: Report submission in week 5

- Maximum number of words: 1500
- Must be submitted online on Moodle
- The report's file extension must be doc or Docx

Assessment topic: Intelligent Agents

Task details:

This assignment requires students to research and critically analyze the concept of intelligent agents and their application in solving complex problems. Students will select a problem domain (e.g., pathfinding, decision-making, or optimization) and propose how an intelligent agent can be designed to address the challenges in this domain. The focus will be on theoretical understanding, analysis, and evaluation rather than implementation.

The assignment includes a detailed report, which must cover the agent's architecture, environment interaction, decision-making strategies, and an analysis of the chosen domain.

Students are tasked with conducting a comprehensive study on intelligent agents and their application to a specific problem domain. The primary goal is to demonstrate your understanding of intelligent agent architectures, their interactions with different types of environments, and how they can be adapted to solve real-world problems.

Steps for the Assignment:

1. Choose a Problem Domain:
    - Select a domain where intelligent agents can be applied, such as autonomous navigation, game playing, resource management, or optimization problems like the N-Queens problem.
2. Research and Analysis:
    - Research existing intelligent agent solutions for the chosen domain.
    - Propose a theoretical intelligent agent design to address the problem, choosing an appropriate agent architecture (e.g., reflex-based, goal-based, utility-based).
    - Analyze the decision-making process and its interaction with the environment.
3. Evaluate the Agent:
    - Discuss the expected performance of the proposed agent in the given problem domain.
    - Analyse the complexity of the agent's design and decision-making process, including time and space complexity.
    - Explore potential challenges and limitations in applying the agent to the domain.
4. Report Requirements:

- Introduction: Introduce the problem domain and the role of intelligent agents in addressing its challenges.
- Agent Design: Explain the proposed agent's architecture and decision-making strategies.
- Environment and Interaction: Describe the environment in which the agent operates and how it interacts with it.
- Analysis:
  - Evaluate the proposed agent's strengths and weaknesses.
  - Discuss the complexity of the approach (time and space).
  - Suggest improvements or alternative approaches.
- Conclusion: Summarize the findings and potential implications for the chosen domain.

Assessment 3 Marking Rubric (20%)

Criteria	**Fail (0-49%)**	**Pass (50-64%)**	**Credit (65-74%)**	**Distinction (75-84%)**	**High Distinction (85-100%)**
Problem Domain Analysis (7%)	The problem domain is poorly analysed or irrelevant. Proposed agent design is vague or inappropriate. No justification for chosen architecture or strategies.	Provides a basic analysis of the problem domain. Proposed agent design is described but lacks depth. Limited justification for chosen architecture or strategies.	Provides a clear and relevant analysis of the problem domain. Proposed agent design is well-described with some justification for chosen architecture and strategies.	Provides a comprehensive analysis of the problem domain. Proposed agent design is detailed and well-justified, with strong links to chosen architecture and strategies.	Provides an outstanding and innovative analysis of the problem domain. Proposed agent design is exceptionally detailed and well-justified, with insightful links to architecture and strategies.
Complexity Analysis (7%)	Complexity analysis is missing, incorrect, or superficial. No meaningful evaluation of time or space complexity.	Provides a basic complexity analysis with limited evaluation of time and space complexity. May include errors or lack detail.	Provides a sound complexity analysis with accurate evaluation of time and space complexity. Demonstrates understanding of computational constraints.	Provides a detailed and accurate complexity analysis. Evaluation of time and space complexity is thorough and well-supported by reasoning.	Provides an exceptional complexity analysis. Evaluation of time and space complexity is highly detailed, accurate, and supported by strong reasoning and critical insights.
Report Quality (6%)	The report is poorly organised, unclear, or lacks depth. Does not meet word count requirements. Contains significant errors in	The report is organised but lacks clarity or depth in some sections. Meets basic word count requirements but contains minor language	The report is well-organised and clear, with adequate depth. Meets word count requirements and has minimal language or formatting errors.	The report is highly organised and clear, with significant depth in analysis. Meets word count requirements and is professionally written and formatted.	The report is exceptionally well-organised, clear, and insightful. Demonstrates outstanding depth in analysis. Professionally written and formatted, exceeding expectations.

	language or formatting.	or formatting errors.			
Total Mark: 20%	COMMENTS:				

Assessment 4

Assessment Type: Individual Project(1500 Words)

Purpose: The primary objective of this assignment is to implement a search strategy on a benchmarking problem and be able to analyse the complexity. This assessment contributes to the learning outcomes a, b, c, and d.

Value: 30% **Due Date:** Weeks 8 and 9

Submission:

1. Program Code: Submit your source code in a zipped folder. Include instructions on how to run your program and any dependencies required. Due week 8
2. Report: Submit a professionally formatted PDF report. Due week 8
3. Demonstration: Attend the scheduled demonstration session to present your solution. **Your demonstration must be in class unless you enroll in an online class. D**ue week 9.

Assessment topic: Search Strategies

Task Details:
The 8-Puzzle consists of a 3x3 board with 8 numbered tiles and one empty space. The goal is to slide the tiles into the correct position (typically a sequential arrangement from 1 to 8 with the blank tile in the bottom-right corner).. This problem serves as an excellent testbed for search algorithms, offering a blend of simplicity and computational complexity.

In this assignment, you will solve the problem using a search strategy of your choice, implement your solution programmatically, analyze the complexity of your chosen approach, and visualize the search process. Your work will involve writing a report, developing a program, and presenting your solution.

The assignment consists of three major components: program implementation, a comprehensive report, and a demonstration.

Useful video: https://www.youtube.com/watch?v=EPgALpUMUhU

1. Program Implementation

You will write a program to solve the 8-Puzzle Problem using a search algorithm. You are free to choose any programming language you are comfortable with. You can choose any search algorithm to solve this problem. Your implementation should be flexible enough to handle different initial board configurations, and should be able to determine whether a solution is possible. The program must be functional, efficient, and well-documented with comments explaining the key parts of your code.

2. Report Submission

The report is an essential part of this assignment, serving as a detailed explanation of your approach. It must include:

- A clear problem statement that explains the problem, its constraints, and its relevance in the context of intelligent agents and search algorithms.
- A discussion of the search strategy you selected, including the reasoning behind your choice. Justify why this search algorithm is appropriate for solving the problem.
- A detailed explanation of your algorithm, including its working mechanism, any specific optimisations or enhancements you applied, and examples where relevant. Use diagrams or flowcharts to make your explanation clear and accessible.
- An analysis of the time complexity and space complexity of your chosen search strategy. Explain the trade-offs involved in your approach and discuss its limitations or potential improvements.
- A visualization of the search process. This visualization should demonstrate how your algorithm progresses towards finding a solution. For example, you can create a step-by-step graphical representation showing the placement of queens on the board during the search.
- Results and observations based on running your program. Discuss the performance of your algorithm, highlighting its strengths and weaknesses.

The report should be professionally written, clear, and concise, with a recommended length of 1500 words.

3. Demonstration

You will be required to demonstrate your program to the instructor or peers. During the demonstration, you should:

- Show how your program solves the problem.
- Explain the working of your algorithm and how it handles constraints.
- Present your visualizations and highlight key aspects of the search process.
- Be prepared to answer questions about your implementation and analysis.

Assessment 4 Marking Rubric (30%)

Criteria	**Fail (0-49%)**	**Pass (50-64%)**	**Credit (65-74%)**	**Distinction (75-84%)**	**High Distinction (85-100%)**
Understanding of the Problem (2%)	Demonstrates little or no understanding of the problem. Incomplete or unclear explanation of the problem and its constraints.	Basic understanding of the problem. Explanation is present but lacks clarity or depth.	Clear explanation of the problem, with a reasonable understanding of its constraints and relevance.	Demonstrates a deep understanding of the problem with clear, concise explanation, including its constraints and importance in AI.	Provides an exceptional explanation with insightful details about the problem, its constraints, and broader relevance to AI and search strategies.
Algorithm Design and Implementation (7%)	Algorithms are either not implemented or do not function as required. Major issues in logic or functionality.	Algorithm functions but may have inefficiencies or errors. Basic implementation of the search strategy.	The algorithm is implemented correctly with minor inefficiencies. Clear application of the chosen search strategy.	Well-implemented and efficient algorithm with thoughtful design. Demonstrates a clear understanding of the search strategy and its application to the problem.	Exceptionally designed and implemented algorithm, demonstrating innovative approaches or optimizations. Code is highly efficient and effectively solves the problem.

Criteria	Fail (0-49%)	Pass (50-64%)	Credit (65-74%)	Distinction (75-84%)	High Distinction (85-100%)
Complexity Analysis (7%)	Little to no analysis of time and space complexity. The explanation is missing or unclear.	Basic analysis of time and space complexity. Explanation may lack depth or precision.	Clear and accurate analysis of time and space complexity with reasonable justification of the results.	Comprehensive and insightful analysis of time and space complexity, demonstrating critical understanding and awareness of algorithm trade-offs.	Exceptional analysis with in-depth evaluation of complexity. Provides clear justifications and insightful observations regarding the algorithm's efficiency and limitations.
Visualization and Presentation (7%)	Visualization is missing or irrelevant. The presentation lacks clarity and does not convey the algorithm's convergence effectively.	Basic visualization of the algorithm's convergence is provided, but lacks depth or clarity. Presentation is adequate but could be improved.	Clear visualization of the algorithm's convergence with appropriate representation. The presentation is clear and effectively communicates the approach.	Detailed and professional visualization effectively demonstrating the algorithm's convergence. The presentation is engaging and demonstrates clear understanding of the methodology.	Exceptional visualization that clearly illustrates the algorithm's convergence with insightful observations. The presentation is highly engaging, professional, and demonstrates deep understanding.
Demonstration (7%)	Unable to explain or demonstrate the implementation effectively. Lacks clarity or preparation.	Demonstration is basic and covers the functionality, but lacks depth or confidence.	Clear and effective demonstration of the algorithm. Reasonable explanation of the implementation and the results.	Demonstrates a strong understanding of the implementation. Provides clear and confident explanations and answers to questions about the solution.	Exceptional demonstration showcasing a deep understanding of the implementation. Explains the approach and results with clarity, confidence, and strong technical insights.
Total Mark: **30%**	COMMENTS: The assessment rubric has a demonstration. The requirements of the completion of this assessment looks for a demonstration to be completed by the students.				

Assessment 5

Assessment Type: Knowledge based Schemes - group project

Assessment topic: Implementation of knowledge-based schemes on a benchmark dataset

Purpose: Students will be working in a group of 4 students. Each student is required to work towards this project. This assessment is designed to reinforce the foundation theories taught and develop students' skills and application of knowledge of the subject content to provide useful insights for businesses. This assessment relates to learning outcomes a, b, c, and d.

Value: 40% **Due Date:** Week 11 for report and demonstration in week 12

Submission: Report submission on Moodle in week 11 and demonstration in weeks 12.**Your demonstration must be in class unless you enroll in an online class.**

Task Details:

You can choose any dataset available on Kaggle and consult with their tutor on how they want to apply machine learning models to extract hidden knowledge. Upon the tutor approval, students in the form of a group of up to 4 members will continue to address the following criteria.Download the dataset from here: https://www.kaggle.com/datasets?tags=13302-Classification

Students are recommended to use the following report structure that address the marking rubric criteria:

- Introduction: introduces the case study and the objective
- Data understanding: Visualise the data. Explain the data. What preparation methods are required? What feature plays an important role in your prediction model?
- Model implementation and evaluation: show the screenshot of your implemented models followed by the explanation. How would you improve the performance of your models? Compared the models against each other.
- Insights: Discuss the output that you receive for the implemented models. Explain the knowledge and insight about the examples in the dataset.
- ACS code of professional ethics: Unfortunately, the machine learning system's decision-making process was not transparent, making it difficult to determine the rationale behind its decisions. How would you assist the officers in this jurisdiction in evaluating whether to use this system as a substitute for human judges? To uphold ACS values, what specific considerations from the Code of Professional Ethics should officers be mindful of? ACS Code of Professional Ethics
- Conclusion: Conclude the model implementation and the results you receive from them. Highlight the key points in the insights.
- References: follow a consistent format and use Harvard style for referencing

Insert the rapid miner file into your report word file and submit your word file on Moodle.

Students will create a process in Rapid Miner generating knowledge from the dataset.You may use different techniques to produce insights about the dataset and conclude with useful advice about the business. Students need to use different visualization techniques to analyse and show the space of the problem. Students are expected to implement a minimum of 4 techniques. Beside the report, students should upload a single process file that includes all the 4 techniques, with a name of the form [StudentID]..

The final report should include:

- Introduction: explaining the business and dataset that you want to work on
- Motivation: you need to highlight the motivation behind the use of AI techniques in extracting hidden knowledge

- Model Implementation: You need to discuss the 4 models implemented. The outcomes of the models should be comprehensively discussed and analysed.
- Conclusion: You need to conclude the findings of this project.

Appendix:
Each group will provide a group work report on how the group has worked together to produce the final report. Each group documents how members discuss, and division of responsibilities and describes how the individual efforts capitalised on the strengths of each group member. It will be used as evidence of individual contributions in the group. It is therefore in each member's interest to ensure that their contribution to the final report is complete. Each group is expected to have at least 3 group meetings for the group assignment. The minutes of group meetings should be documented and attached as an appendix of this group work report, clearly indicating who was present, issues and actions, agreed timelines, and the like. The group work report must indicate that a fair and reasonable distribution of work amongst group members was achieved. Periodic information such as emails or diary entries must be inserted into the correct section in chronological order.

If the submitted group work report suggests that not all contributions were of equivalent standard and effort, differential marks will be awarded to individuals within the same group. It must also document what individual group members understood as their allocated tasks, and that individual group members submitted allocated work of acceptable standard and quality by the date that was agreed upon.

Marking Rubric for Assessment: 5 **Value: 40%**

Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Rapid Miner Source code 6%	Performs less than 50% of task., structure not clear.	Performs 50%-64% of tasks. structure lacking in some parts	Performs 65%-74% of tasks. structure not totally clear	Performs 75%-84% of tasks. Well structured.	Performs more than 85% of task. Very clearly structured
Data Understanding and Preparation 6%	Unsatisfactory data preparation	Satisfactory data preparation but no filtering, no fair distribution of samples	Good data preparation but lack of filtering	Excellent data preparation But no proper set of attributes is selected or there are outliers	Exceptional data preparation. A variety of feature selections are tested. Exceptional feature extraction is done selected features are combinations of existing and extracted features
Model Implementation and performance analysis 6%	Unsatisfactory intelligent models accuracy less than 60%	Satisfactory intelligent models are implemented but the accuracy is poor or there is no proper objective. accuracy above 60%	Effective intelligent models with a fair accuracy. The parameters could be better adjusted accuracy above 75%	Excellent intelligent models With good accuracy. Still there is a chance of better results accuracy above 85%	Exceptional intelligent models with excellent outcome Accuracy above 90%

Draw a conclusion based on the insights and analysis 6%	No conclusion presented and no visualisation	Conclusion based on some argument and satisfactory visualisation	Conclusion based on sound argument and effective visualisation	Conclusion based on an extensive argument and excellent visualisation	Conclusion based on an extensive and compelling argument and exceptional visualisation
ACS code of professional ethics 6%	The system's decision-making process is completely opaque with no means to understand or explain how decisions are made, failing to adhere to ethical standards of transparency and accountability.	The system provides minimal insight into the decision-making process, offering only basic explanations that partially meet ethical standards of transparency and accountability.	The system offers a reasonable level of insight into the decision-making logic, providing explanations that are generally clear but may lack depth, meeting most ethical standards of transparency and accountability.	The system provides clear and detailed explanations of its decision-making process, effectively meeting ethical standards of transparency and accountability with minor areas for improvement.	The system offers comprehensive, highly transparent explanations of its decision-making process, fully meeting and exceeding ethical standards of transparency and accountability as outlined in the ACS Code of Professional Ethics.
Group Demonstration 5%	Very little subject matter knowledge shown.	Adequate subject matter knowledge shown.	Sound subject matter knowledge shown.	High level of subject matter knowledge shown.	High level of subject matter knowledge and synthesis with theory shown
Individual Demonstration 5%	Very little subject matter knowledge shown.	Adequate subject matter knowledge shown.	Sound subject matter knowledge shown.	High level of subject matter knowledge shown.	High level of subject matter knowledge and synthesis with theory shown
Total Mark: / 40%	**COMMENTS:** The assessment rubric has a demonstration. The requirements of the completion of this assessment looks for a demonstration to be completed by the students.				

3.2 General information about assessment

a) Late Penalties and Extensions

An important part of business life and key to achieving KOI's graduate outcome of Professional Skills is the ability to manage workloads and meet deadlines. Completing assessment tasks on time is a good way to master these habits.

Students who miss mid-trimester tests and final exams without a valid and accepted reason may not be granted a deferred exam and will be awarded 0 marks for the assessment item. Assessment items which are missed or submitted after the due date/time will attract a penalty unless there is a compelling reason (see below). These penalties are designed to encourage students to develop good time management practices, and to create equity for all students.

Any penalties applied will only be up to the maximum marks available for the specific piece of assessment attracting the penalty.

Late penalties, granting of extensions and deferred exams are based on the following:

In Class Tests and Quizzes (excluding Mid-Trimester Tests)

- Generally, extensions are not permitted. A make-up test may only be permitted under very special circumstances where acceptable supporting evidence of illness, hardship or unavoidable problems preventing completion of the assessment is provided (see section (b) below). The procedures and timing to apply for a make-up test (only if available) are as shown in the section *Applying for an Extension* (see below).
- Missing a class test will result in 0 marks for that assessment item unless the above applies.

Written Assessments and Video Assessments

- There is a late penalty of *5% of the total available marks* per calendar day unless an extension is approved (see *Applying for an Extension section* below).

Presentations

- Generally, extensions are not permitted. Missing a presentation will result in 0 marks for that assessment item. The rules for make-up presentations are the same as for missing in-class tests (described above).
  For group presentations, if serious circumstances prevent some members of the group from participating, the members of the group who are present should make their contributions as agreed. If a make-up presentation is approved, the other members of the group will be able to make their individual presentation later and will be marked according to the marking rubric. A video presentation may be used to facilitate the process.

Mid-Trimester Tests and Final Exams

If students are unable to attend mid-trimester tests or final exams due to illness, hardship or some other unavoidable problem (acceptable to KOI), they must:

- Complete the Assignment Extension / Exam Deferment Form available by contacting academic@koi.edu.au as soon as possible*, but no later than three (3) working days after the exam date.*
- Provide acceptable documentary evidence (see section (b) below).
- Agree to attend the deferred exam as set by KOI if a deferred exam is approved.

Deferred exam

- There will only be one deferred exam offered.
- Marks obtained for the deferred exam will be the marks awarded for that assessment item.
- If you miss the deferred exam you will be awarded 0 marks for the assessment item. This may mean you are unable to pass the subject.

b) Applying for an Extension

If students are unable to submit or attend an assessment when due, they must

- Complete the Assignment Extension / Exam Deferment Form available by contacting academic@koi.edu.au as soon as possible*, but no later than three (3) working days of the assessment due date.*

- Provide acceptable documentary evidence in the form of a medical certificate, police report or some other appropriate evidence of illness or hardship, or a technician's report on problems with computer or communications technology, or a signed and witnessed statutory declaration explaining the circumstances.
- Students and lecturers / tutors will be advised of the outcome of the extension request as soon as practicable.

Please remember there is no guarantee of an extension being granted, and poor organisation is not a satisfactory reason to be granted an extension.

c) Referencing and Plagiarism

Please remember that all sources used in assessment tasks must be suitably referenced. Failure to acknowledge sources is plagiarism, and as such is a very serious academic issue. Students plagiarising run the risk of severe penalties ranging from a reduction in marks through to 0 marks for a first offence for a single assessment task, to exclusion from KOI in the most serious repeat cases. Exclusion has serious visa implications. The easiest way to avoid plagiarising is to reference all sources.

Harvard referencing is the required method – in-text referencing using Author's Surname (family name) and year of publication. A Referencing Guide, "Harvard Referencing", and a Referencing Tutorial can be found on the right-hand menu strip in Moodle on all subject pages.

An effective way to reference correctly is to use *Microsoft Word's* referencing function (please note that other versions and programs are likely to be different). To use the referencing function, click on the References Tab in the menu ribbon – students should choose *Harvard.*

Authorship is also an issue under plagiarism – KOI expects students to submit their own original work in both assessment and exams, or the original work of their group in the case of a group project. All students agree to a statement of authorship when submitting assessments online via Moodle, stating that the work submitted is their own original work.

The following are examples of academic misconduct and can attract severe penalties:

- Handing in work created by someone else (without acknowledgement), whether copied from another student, written by someone else, or from any published or electronic source, is fraud, and falls under the general Plagiarism guidelines.
- Copying / cheating in tests and exams is academic misconduct. Such incidents will be treated just as seriously as other forms of plagiarism.
- Students who willingly allow another student to copy their work in any assessment may be considered to assist in copying/cheating, and similar penalties may be applied.

Where a subject coordinator considers that a student might have engaged in academic misconduct, KOI may require the student to undertake an additional oral exam as a part of the assessment for the subject, as a way of testing the student's understanding of their work.

Further information can be found on the KOI website.

d) Reasonable Adjustment

The Commonwealth Disability Discrimination Act (1992) makes it unlawful to treat people with a disability less fairly than people without a disability. In the context of this subject, the principle of Reasonable Adjustment is applied to ensure that participants with a disability have equitable access to all aspects of the learning for the subject. For assessment, this means that barriers to their demonstrating competence are removed wherever it is reasonably practical to do so.

Examples of reasonable adjustment in assessment may include:

- provision of an oral assessment, rather than a written assessment
- provision of extra time
- use of adaptive technology.

The focus of the adjusted assessment should be on enabling the student to demonstrate achievement of the learning outcomes for the subject, rather than on the method of assessment.

e) Appeals Process

Full details of the KOI *Assessment and Assessment Appeals Policy* may be obtained in hard copy from the Library, and on the KOI website www.koi.edu.au under *Policies and Forms*.

Assessments and Mid-Trimester Exams:

Where students are not satisfied with the results of an assessment, including mid-trimester exams, they have the right to appeal. The process is as follows:

- Discuss the assessment with their tutor or lecturer – students should identify where they feel more marks should have been awarded – students should provide valid reasons based on the marking guide provided for the assessment. Reasons such as *"I worked really hard*" are not considered valid.
- If still not satisfied, students should complete an Application for Review of Assessment Marks form, clearly explaining the reasons for seeking a review. This form is available from the KOI website under *Policies and Forms* and is also available at KOI Reception (Kent St, Market St and O'Connell St). The completed Application for Review of Assessment Marks form should be submitted as explained on the form with supporting evidence attached to academic@koi.edu.au .
- The form must be submitted within *ten (10) working days* of the return of the marked assessment, or within *five (5) working days* after the return of the assessment if the assessment is returned after the end of the trimester.

Review of Grade – whole of subject and final exams:

Where students are not satisfied with the results of the whole subject or with their final exam results, they have the right to request a Review of Grade – see the *Assessment and Assessment Appeals Policy* for more information.

An *Application for Review of Grade/Assessment Form* (available from the KOI Website under *Policies and Forms* and from KOI Reception at Kent St, Market St and O'Connell St) should be completed clearly explaining the grounds for the application. The completed application should be submitted as explained on the form, with supporting evidence attached to academic@koi.edu.au.